Curriculum vitae
Giancarlo Croce

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Research Interests

My research sits at the interface of biology, artificial intelligence, and statistical modelling to tackle fundamental questions in immunology. I develop data-driven approaches to predict T- and B-cell receptor - antigen interactions. Previously, I developed computational tools to model protein evolution and protein–protein interactions.

Professional experience

SEP 2025 - CURRENT

Machine learning scientist at IMPRINT - a non-profit Focused Research Organizations (FRO) dedicated to decoding the immune system

JAN 2020 - JUN 2025
LAUSANNE (SWITZERLAND)

Postdoctoral fellow, *University of Lausanne - Ludwig center for cancer research*
Marie Skłodowska-Curie Individual Fellowship MT-PoINT project
Cancer Immune Recognition Lab - David Gfeller Lab

Education

OCT 2016 - OCT 2019,
PARIS (FRANCE)

PhD, Computational biology, *Sorbonne University*
Computational and Quantitative Biology Lab
Thesis: Towards a genome-scale coevolutionary analysis
Supervisor: Martin Weigt. Co-supervisor: Olivier Tenaillon

SEP 2014 - JUN 2016,
PARIS (FRANCE)

MSc in theoretical and statistical physics, *École Normale Supérieure*
Master thesis: *Out of equilibrium dynamics of spin glass models*
Supervisor: Cristina Toninelli

SEP 2011 - JUL 2014,
PAVIA (ITALY)

BSc in physics, *University of Pavia*
Thesis: *Symmetries in quantum mechanics, Casimir Operators*
Supervisor: Prof. Claudio Dappiaggi

Grants and awards

Sep 2025 Awarded the Remarkable Output distinction by the Swiss Bioinformatics Society.

Jun 2024 - Selected to participate in the 73rd Lindau Nobel Laureate Meeting, engaging with Nobel Laureates and leading scientists.

Jul 2021 - Jul 2023 Marie Sklodowska-Curie Individual Fellowship 2020 for the MT-PoINT (Motif in T cells for the Prediction of INTeractions) project at the interface between data science and immunology

Jan 2019 - Mar 2019 Visiting researcher at University of L'Havana (Cuba) funded by the INFERNET project: new algorithms for inference and optimization from large-scale biological data

Oct 2016 - Nov 2019 Ph.D. supporting grant funded by *Labex Calsimlab - Data,*

computing and simulation.

Sep 2014 - Jun 2016 Two-year master's scholarship for international students from the École Normale Supérieure, Paris.

Sep 2011 - Sep 2014 Student at Collegio Ghislieri, a 450-year-old Italian institution providing housing and financial support to students enrolled at the University of Pavia.

Teaching experience

Courses:
Fall 2024 - Write a review - course for master students
Sep-Nov 2022/2023/2024 - Introduction to clinical bioinformatics - for undergraduate students
Dec 2021 - Introduction to Cancer Genomics - for PhD students

Student supervisor:
Master's thesis: Jikaël Ntoko, Daniel Hafez, Richie Yat-Tasi Wan
Master's and Bachelor's Projects: Alice Chiozza, Cailyn Mae Ong, Yebeen Kim, Animesh Awasth

Recent conferences and talks

Jun 2026 - Meldola (Italy)
1st Immunorepertoire and biologics development conference

Nov 2025 - New York (USA)
Biohub Affiliate Symposium - Chan Zuckerberg Initiative

Sep 2025 - Basel (Switzerland)
Basel Computational Biology Conference — Bioinformatics meets AI. Awarded the SIB Remarkable Output distinction.

May 2025 - Antwerp (Belgium)
The Fourth Antwerp TCR meeting - international experts discuss T cells from the TCR perspective.

Jan 2025 - University of Cologne (Germany)
Predictability in Evolution - Collaborative Research Center seminal series

Reviewing activities

Reviewer for Nature Communications, Bioinformatics, Cell Systems, Cell Genomics, Immunoinformatics, Plos Computational Biology, Plos ONE

Languages

- **Italian**: Native speaker
- **English**: Fluent
- **French**: Fluent